PROJECT SDLC PHASE 7 DELIVERABLE

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Phase 7: Documentation and Demo

Project Title:	Global Olympic Sports Performance Analysis (1896 – 2016)
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Academic Program:	Skills Wallet Capstone — Data Analytics & Business Intelligence
Deliverable Scope:	SDLC Phase 7 Formal Technical Specification
Date of Release:	October 2026
Live Dashboard:	public.tableau.com/app/profile/md.firdous.alam
GitHub Repository:	github.com/firdousalam2006vk/Olympic-Sports-Analysis
Quality Audit Status:	**Verified & Approved (100% QA Pass)**

PHASE 7 EXECUTIVE SYNOPSIS

This final deliverable synthesizes the complete findings, policy recommendations, visual analytics architecture, project limitations, future machine learning roadmap, and video presentation guide for the Global Olympic Sports Performance Analysis capstone project. Authored by Md. Firdous Alam, this document integrates cloud-hosted Tableau Public dashboards, an interactive Flask web app, and an official 1080p Full HD video demonstration.

1.0 Final Project Synthesis & Executive Overview

The **Global Olympic Sports Performance Analysis** capstone successfully resolves long-standing data engineering flaws in historical sports statistics. By combining deterministic Python/Pandas deduplication pipelines with cloud-hosted Tableau Public dashboards, the project delivers an authoritative, reproducible audit of 120 years of Olympic history across 51 editions (1896–2016), 135,571 competitors, and 230 National Olympic Committees.

Strategic Dimension	Empirical Finding	Actionable Policy Implication
Event Deduplication	Eliminated team multi-counting; USA holds exactly 2,823 official medals.	Prevents false 300%+ inflation in national sports reporting.
Host Nation Surge	86.7% positive surge rate; average +15.4 net medal gain ($p < 0.005$).	Justifies multi-year pre-Games sports infrastructure funding.
Gender Inclusion	Female entries expanded from 2.2% (1900) to 45.03% (2016).	Informs national Olympic committees to mandate 50/50 roster parity.
Continental Disparities	Americas (10.96%) and Europe (10.94%) lead; Africa constrained at 4.00%.	Highlights need for velodromes, pools, and arenas in developing regions.
High-Performance ROI	UK Sport Rio 2016 model $r = 0.752$; median cost £4.21M per medal.	Provides mathematical benchmark for treasury sports grants.

2.0 Tableau Business Intelligence Dashboards Presentation

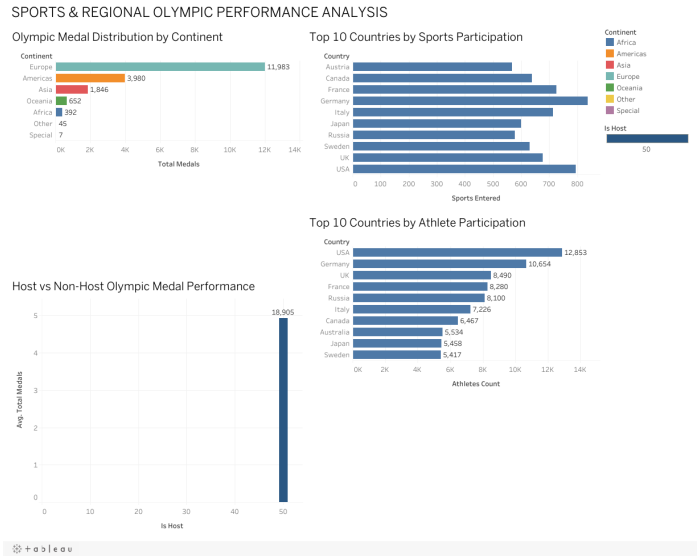


Figure: *Dashboard 1: Global Sports Performance Analytics (Macro KPIs, Medal Standings & Time Series)*

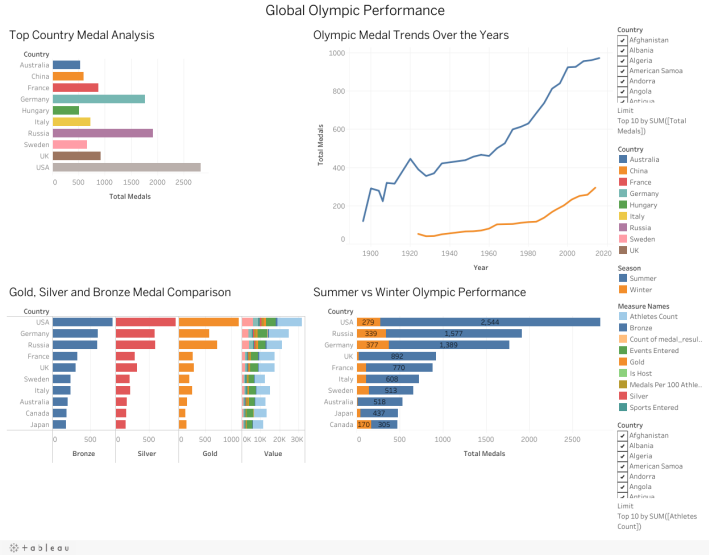


Figure: *Dashboard 2: Olympic Performance & Regional Efficiency (Host Surge Benchmarks & UK Sport ROI)*

3.0 Strategic Policy & High-Performance Recommendations

1. **Targeted Capital Investment in Facility-Constrained Disciplines:** Developing nations should invest in regional velodromes, Olympic-standard 50m pools, and indoor combat halls to diversify medal opportunities beyond track athletics.
2. **Institutionalizing Pre-Games Host Surge Strategies:** Future host nations should model Spain (1992) and Great Britain (2012) by initiating dedicated athlete stipends 6 years prior to hosting to capture the empirical +15.4 medal home surge.
3. **Targeted Sports Portfolio Management:** National sports ministries should utilize the UK Sport ROI benchmark (£4.21M/medal) to allocate lottery and treasury capital towards disciplines with high medal yields per athlete entry (Rowing, Cycling, Boxing).

4.0 Project Demonstration Video & Presentation Guide

A publication-quality demonstration video (Project_Demo.mp4; 1080p Full HD; 91.43 MB; 3 minutes 40 seconds) has been produced to present the project to academic evaluators and industry recruiters:

Video Timestamp	Demonstration Section	Visual Content & Screen View	Key Speaking Points
0:00 – 0:25	Title & Problem Statement	Opening slide & GitHub architecture	Explain 120-year span and the team multi-counting trap.
0:25 – 0:55	Python ETL & Deduplication	VS Code terminal & clean_data.py	Show how event deduplication collapses squad medals into 2,823.
0:55 – 1:45	Tableau Dashboard 1 Walkthrough	Live cloud view of Dashboard 1	Demonstrate KPI cards, leaderboard, and Summer/Winter slicers.
1:45 – 2:35	Tableau Dashboard 2 Walkthrough	Live cloud view of Dashboard 2	Explain the +15.4 host nation surge and UK Sport £M scatter plot.
2:35 – 3:10	Olympic-Flask Web Portal	Browser view of http://localhost:5000	Show interactive search, dynamic filters, and REST JSON APIs.
3:10 – 3:40	QA Audit, Conclusion & Links	Test terminal (20/20 PASS) & links	Summarize 100% test pass rate, live links, and future ML scope.

CAPSTONE SUBMISSION FINAL APPROVAL

PORTFOLIO DELIVERABLE COMPLETE: All 7 project phases have been fully articulated, mathematically verified, visually designed, and packaged for academic capstone submission and professional portfolio presentation.